

Spring AI: Capstone Project

Enterprise Multi-Agent Knowledge & Operations Assistant

**The Pinnacle of Enterprise Generative AI in Modern
Java**

1. Capstone Overview & Architecture

- The Capstone Project synthesizes all 12 modules into a unified, production-grade enterprise platform:
 - **Supervisor Agent Architecture:** Intelligent intent triaging and subagent coordination.
 - **Knowledge RAG Subagent:** Enterprise policy and contract retrieval with strict grounding.
 - **Operations Subagent:** IT service management, automated ticketing, and server diagnostics.
 - **Zero-Trust Guardrails:** Inbound PII redaction and multi-turn safety constraints.

2. Multi-Agent Orchestration Flow

- **Step 1: Input Ingestion & Redaction:**

- Employee prompts enter via `/api/v1/assistant/chat`.
- Guardrail filters scrub credit card and sensitive PII tokens before LLM exposure.

- **Step 2: Supervisor Decision Engine:**

- Directs context-heavy queries to `KnowledgeRagSubagent`.
- Routes operational commands to `OperationsToolSubagent`.

- **Step 3: Execution & Audit Aggregation:**

- Combines subagent output into `CapstoneResponse` with complete execution audit trail.

3. High-Performance Spring Boot 3.4 Runtime

- **Project Loom Virtual Threads:**
 - High concurrency with zero blocking thread exhaustion.
- **GraalVM Native AOT Ready:**
 - Pre-registered runtime reflection hints and template bundles.
- **Actuator Health & Prometheus:**
 - Real-time AI subsystem health indicators and latency monitoring.

4. Graduation & Production Roadmap

- You have completed the Spring AI Enterprise Master Course:
 - Foundations, Prompt Engineering, Structured Outputs
 - Tool Calling, Callbacks, Advanced RAG, Multimodal AI
 - Advisors & Memory, Model Context Protocol (MCP)
 - Agentic Multi-Agent Workflows, Production Security & Native AOT
- You are now equipped to design and scale enterprise-grade AI applications in Java!